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Wisdom of the LLM Crowd: A Large Scale Benchmark of Multi-Label U.S. Election-Related Harmful Social Media Content

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Extended Abstract

Online harmful content, including misinformation, increasingly undermines modern elections by shaping public opinion, influencing voter behavior, and destabilizing democratic processes [3, 1, 6, 2]. This challenge is amplified by the growing volume of online content and the increasing reliance on social media platforms for political information, whose networked diffusion mechanisms facilitate the rapid spread of misleading narratives [8]. As online harmful content continues to influence elections, a limited number of standardized datasets (*e.g.,* [15, 7, 4, 10, 9]) have been developed to evaluate election-related harmful content. Many of these efforts are framed using terms such as misinformation and fake news, reflecting prevailing terminology [17], while focusing on specific classes of election-related harms. These developments underscore the pressing need for robust, election-specific datasets to support the development of more effective systems for detecting harmful content. Raw social media data are often messy and unstructured [16, 5], underscoring the importance of curated, cleaned datasets for advancing research on online harmful content. However, developing such datasets poses significant challenges. Annotators must invest substantial time and effort to categorize content manually, a process that is costly and prone to inconsistency, especially when classifying ambiguous or subjective material [11]. Although some researchers have proposed automated annotation using large language models LLMs [14, 13], few have systematically compared the performance or evaluated the accuracy of different LLMs when labeling diverse categories of harmful content. In this study, we evaluate the effectiveness of LLMs in classifying potentially harmful or sensitive online content. And, we examine whether discernible patterns emerge in the use of generative AI as a supportive annotation tool.

Specifically, **(RQ1) How do different LLMs perform in annotating online harmful content, and which model characteristics explain their annotation behavior?** We collected nearly 100k posts from X.com between October 17, 2024, and July 16, 2025, covering the U.S. presidential election period, and evaluated six leading LLMs with different architectures that may contain potentially harmful information across five categories: Conspiracy, Hate Speech, Satire, Sensationalism, and Speculation. As shown in Figure 1, we validate a 1% subset [12] via human annotation (n = 34) on a crowdsourcing platform and adopt a wisdom of the crowd

approach across the three best-performing LLMs, using majority voting to establish consensus labels. Figure 2 compares six LLMs and human annotation (majority vote) based on annotation counts across five categories and their distributions. Results indicate that LLMs generally detect a high proportion of post content as Speculation (ranging from 32.8% - 58.6%), Sensationalism (ranging from 18.8% - 38.7%), and Conspiracy (ranging from 16.7% - 26.9%). On the other hand, the lower categories identified by LLMs are generally Hate Speech (ranging from 4.3% - 8.6%) and Satire (ranging from 3.6% - 7.7%). These distribution patterns are similar to those found in human annotations. Gemini 2.5 Pro consistently has a higher label count across nearly all categories compared to the other models.

(RQ2) How do human annotators perform in annotating online harmful content, and to what extent do human annotators' demographics and characteristics, such as political ideology, shape the annotation of online harmful content? Pearson's χ^2 analyses (Figure 4) revealed statistically detectable but generally small associations between demographics and labeling patterns. Among the five labels, *Speculation* showed the strongest demographic influence, driven largely by political ideology ($\chi^2_{(4)} = 222.49$, $p < 0.001$, $V = 0.27$) and to a lesser extent by political affiliation ($\chi^2_{(3)} = 89.80$, $p < 0.001$, $V = 0.17$). *Hate Speech* was the next most variable across groups, differing by both ideology ($\chi^2_{(4)} = 111.99$, $p < 0.001$, $V = 0.19$) and political affiliation ($\chi^2_{(3)} = 85.76$, $p < 0.001$, $V = 0.17$).

Finally, **(RQ3) To what extent do LLM predictions agree, and how does this agreement compare with human annotators?** We show within-in Krippendorff's α for each LLMs and human agreement in Figure 3. We observed that LLMs achieved higher inter-rater reliability (mean Krippendorff's $\alpha = 0.70$ for optimized model combinations) and more consistency than human annotators (mean $\alpha = 0.62$ for the best annotation sets per category), but with larger variance. With the potential of using LLMs as supporting annotation tools, we first computed Cohen's kappa (κ) for each LLM against the human consensus labels. The results are presented in Table 1. Llama 3.1 (8B) generally performs better across most categories, while Gemini 2.5 Pro shows slightly lower performance. Similar to the human-majority-vote approach, we also applied it across multiple LLMs using the same method. Results, also shown in Table 1, indicate that LLM majority voting generally improves agreement, particularly when combining models from different LLM families. This is an implication where a crowd of three LLMs performs better than a single LLM To further assess direct performance, we report recall, precision, and F1 score in Figure 5, examining how LLM consensus predictions align with human consensus labels. Since annotations in misinformation tasks are often used for detection and typically focus on the positive class, Recall can serve as a more representative metric. The LLMs achieved the highest Recall in Speculation (mean = 0.85 ± 0.03) and Conspiracy (mean = 0.81 ± 0.03), followed by Sensationalism (mean = 0.62 ± 0.07) and Hate Speech (mean = 0.64 ± 0.08). Satire showed the lowest Recall (mean = 0.49 ± 0.08). These results indicate that LLM majority voting aligns well with human consensus for clearly defined categories, while subjective or nuanced categories like Satire remain more challenging.

Overall, this study introduces a large scale and rigorously annotated dataset of U.S. election related social media posts to advance harmful content research. Through a comprehensive crowdsourced annotation effort and systematic benchmarking of six state of the art large language models, we establish a practical and scalable baseline for multi-label harmful content classification. By combining consensus strategies with human oversight, our framework improves upon purely manual or heuristic approaches while maintaining strong performance, achieving a recall of up to 0.90. Majority voting across multiple LLMs parallels established human annotation protocols and substantially reduces time and cost, which is essential for

analyzing the massive volume of content generated daily on online social networks. In addition, we demonstrate that annotator demographic characteristics are associated with labeling tendencies, underscoring the importance of incorporating human factors alongside model calibration in future annotation pipelines. The released dataset further enables investigation of co-occurring harmful content signals, such as hate speech appearing alongside satire, and is publicly available [1] to support ongoing research.

References

[1] Dareen Al-Khoury. What is harmful content? *The Routledge Handbook of Conflict and Peace Communication*, 2024.

[2] Arnav Arora, Preslav Nakov, Momchil Hardalov, Sheikh Muhammad Sarwar, Vibha Nayak, Yoan Dinkov, Dimitrina Zlatkova, Kyle Dent, Ameya Bhatawdekar, Guillaume Bouchard, and Isabelle Augenstein. Detecting harmful content on online platforms: What platforms need vs. where research efforts go. *ACM Comput. Surv.*, 56(3), October 2023.

[3] Cheuk Hang Au, Kevin KW Ho, and Dickson KW Chiu. The role of online misinformation and fake news in ideological polarization: barriers, catalysts, and implications. *Information systems frontiers*, pages 1–24, 2022.

[4] Emily Chen, Ashok Deb, and Emilio Ferrara. # election2020: the first public twitter dataset on the 2020 us presidential election. *Journal of Computational Social Science*, pages 1–18, 2022.

[5] Mohamed Elsayed, Amira Abdelwahab, and Hatem Ahdelkader. A proposed framework for improving analysis of big unstructured data in social media. In *2019 14th International conference on computer engineering and systems (ICCES)*, pages 61–65. IEEE, 2019.

[6] Anastasia Giachanou and Paolo Rosso. The battle against online harmful information: The cases of fake news and hate speech. In *Proceedings of the 29th ACM International Conference on Information & Knowledge Management*, pages 3503–3504, 2020.

[7] Ian Kennedy, Morgan Wack, Andrew Beers, Joseph S Schafer, Isabella Garcia-Camargo, Emma S Spiro, and Kate Starbird. Repeat spreaders and election delegitimization: A comprehensive dataset of misinformation tweets from the 2020 us election. *Journal of Quantitative Description: Digital Media*, 2, 2022.

[8] Patricia L Moravec, Randall K Minas, and Alan R Dennis. Fake news on social media. *MIS quarterly*, 43(4):1343–A13, 2019.

[9] Gabriela Pinto, Charles Bickham, Tanishq Salkar, Joyston Menezes, Luca Luceri, and Emilio Ferrara. Tracking the 2024 us presidential election chatter on tiktok: a public multimodal dataset. In *Companion Proceedings of the ACM on Web Conference 2025*, pages 773–776, 2025.

[1] `https://github.com/Sensify-Lab/USE24-XD`

[10] Shaina Raza, Tahniat Khan, Veronica Chatrath, Drai Paulen-Patterson, Mizanur Rahman, and Oluwanifemi Bamgbose. Fakewatch: a framework for detecting fake news to ensure credible elections. *Social Network Analysis and Mining*, 14(1):142, 2024.

[11] Asaad F Said, Vinay Kashyap, Namrata Choudhury, and Farshad Akhbari. A cost-effective, fast, and robust annotation tool. In *2017 IEEE applied imagery pattern recognition workshop (AIPR)*, pages 1–6. IEEE, 2017.

[12] Hyunjin Song, Petro Tolochko, Jakob-Moritz Eberl, Olga Eisele, Esther Greussing, Tobias Heidenreich, Fabienne Lind, Sebastian Galyga, and Hajo G Boomgaarden. In validations we trust? the impact of imperfect human annotations as a gold standard on the quality of validation of automated content analysis. *Political Communication*, 37(4):550–572, 2020.

[13] Hongjin Su, Jungo Kasai, Chen Henry Wu, Weijia Shi, Tianlu Wang, Jiayi Xin, Rui Zhang, Mari Ostendorf, Luke Zettlemoyer, Noah A Smith, et al. Selective annotation makes language models better few-shot learners. *arXiv preprint arXiv:2209.01975*, 2022.

[14] Zhen Tan, Dawei Li, Song Wang, Alimohammad Beigi, Bohan Jiang, Amrita Bhattacharjee, Mansooreh Karami, Jundong Li, Lu Cheng, and Huan Liu. Large language models for data annotation and synthesis: A survey. *arXiv preprint arXiv:2402.13446*, 2024.

[15] Fatemeh Torabi Asr and Maite Taboada. Big data and quality data for fake news and misinformation detection. *Big data & society*, 6(1):2053951719843310, 2019.

[16] Ivona Velkova. Unstructured social media data processing with artificial intelligence. *Industry 4.0*, 8(2):65–67, 2023.

[17] Jing Zeng and Scott Babwah Brennen. The misinformation. *Internet Policy Review*, 12(4):1725, 2023.

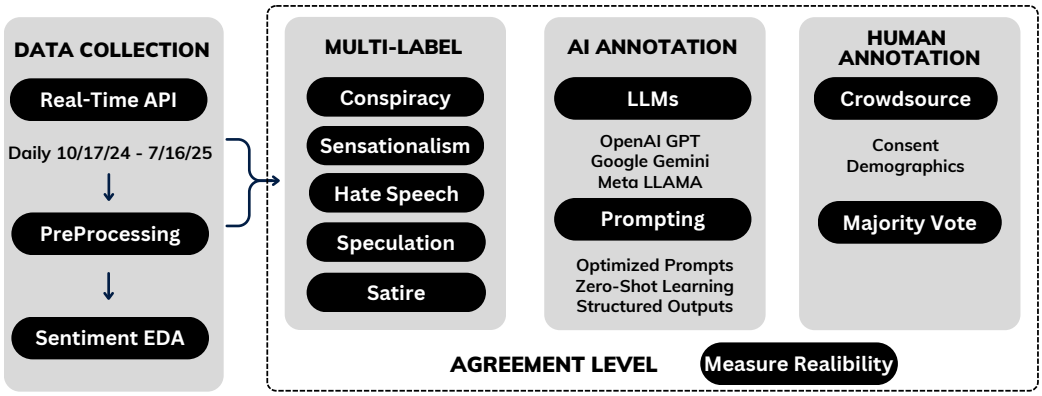


Figure 1: Overall Method for Multi-label Harmful Online Content Annotation

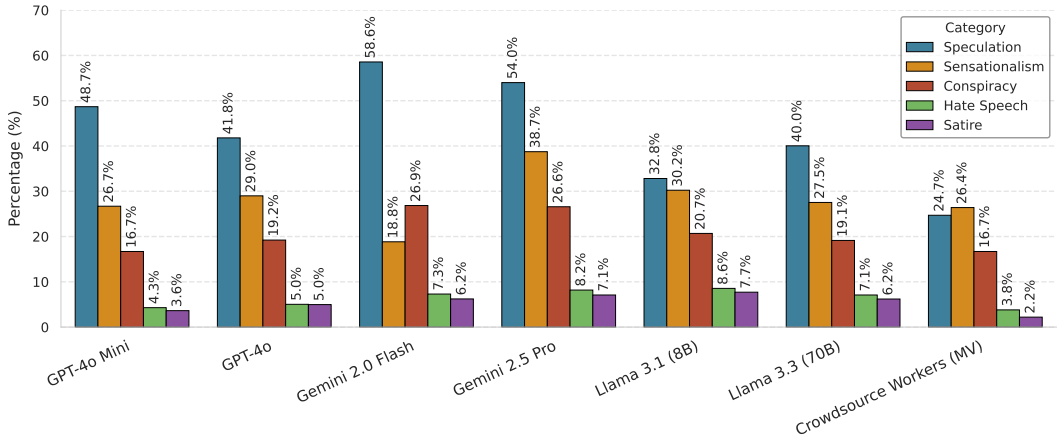
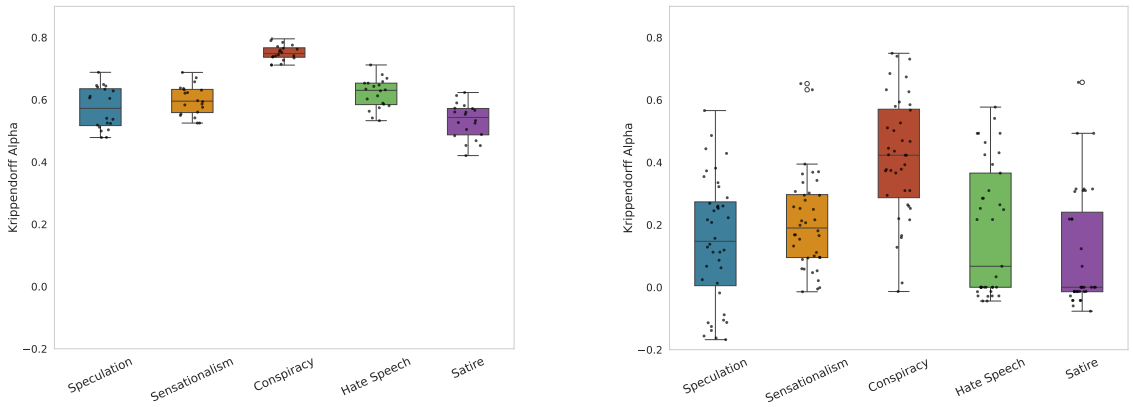


Figure 2: Distribution of each category annotated by different LLM and crowdsourced workers (majority vote). Each category was evaluated independently.



(a)LLMs (n=20) (b)MTurk Workers (n=193)

Figure 3: Krippendorff's Alpha (α) comparison. Figure (a) shows α computed for all combinations of three LLMs selected from six models across the full dataset. Figure (b) presents α for 193 groups of human annotators on the subset of data where annotations were available. Values range from -1 to 1, where higher α indicates stronger agreement.

Table 1: Comparison of Cohen's Kappa (κ) across individual LLM models and the LLM majority vote (MV) from three LLMs, evaluated against crowdsourced workers' consensus labels. The smallest value in each category is highlighted in red, and the largest value is highlighted in green. The last row is reported based on the 15 combination groups for Min and Max of (κ)

Model	**Speculation**	**Sensationalism**	**Conspiracy**	**Hate Speech**	**Satire**
GPT-4o Mini	0.27	0.24	0.60	0.36	0.24
GPT-4o	0.34	0.17	0.62	0.46	0.17
Gemini 2.0 Flash	0.26	0.23	0.60	0.43	0.23
Gemini 2.5 Pro	0.28	0.17	0.56	0.28	0.17
Llama 3.1 (8B)	0.37	0.26	0.60	0.38	0.26
Llama 3.3 (70B)	0.28	0.20	0.59	0.39	0.20
LLMs MV	0.27 - 0.36	0.35 – 0.45	0.60 – 0.65	0.39 - 0.48	0.19 – 0.27

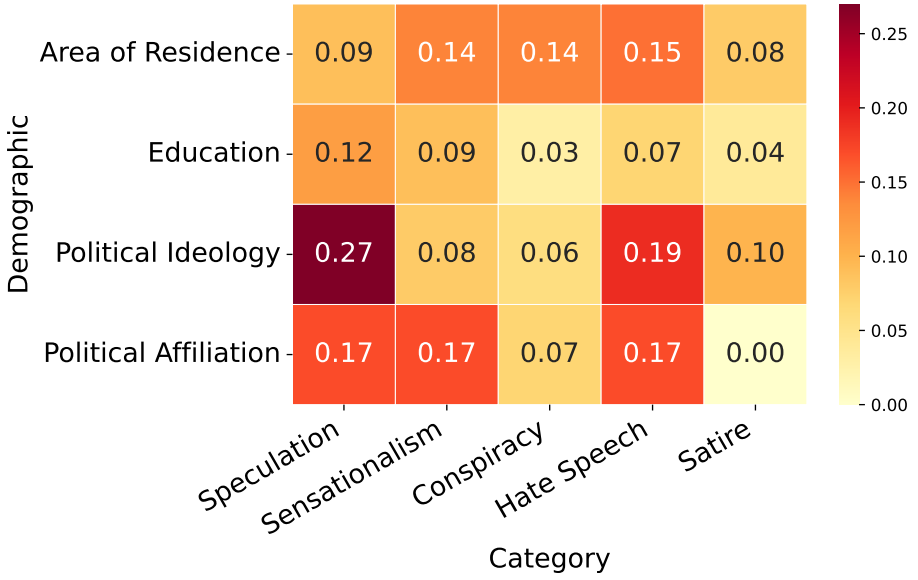


Figure 4: Strength of association between annotators' demographics and annotation categories, expressed as Cramér's V from Pearson's χ^2 tests ($N \approx 3{,}000$ assignments). Darker cells indicate stronger associations.

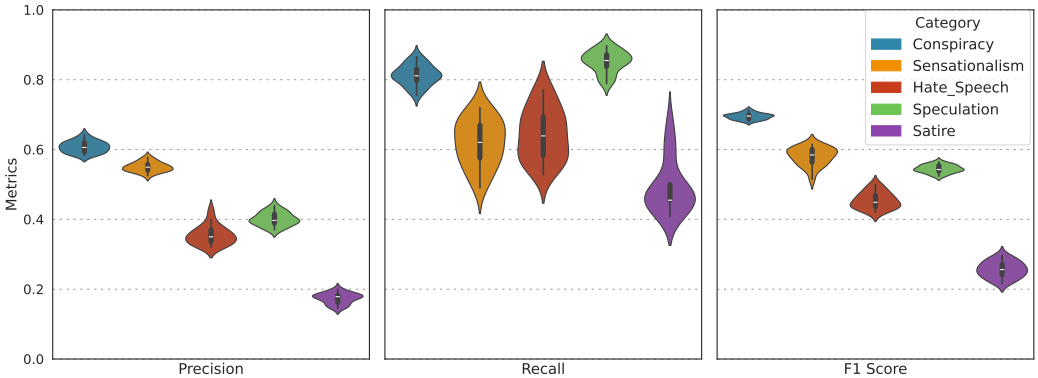


Figure 5: Direct comparison between majority-vote labels produced by LLM ensembles (20 outputs) and majority-vote labels from crowdsourced workers across different categories.